FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: faultsweep
Model status: BOUNDED_DOCUMENTED_FAULT_LOCATION_SWEEP
Captured UTC: 2026-10-26T06:04:42.565Z
Warning: ENGINEERING SCREENING ONLY ? independently verify actual manufacturer, utility, installation, e
quipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certif
ication or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: nominalLLV | Entered value: 13800
Input path: voltageFactor | Entered value: 1.1
Input path: frequencyHz | Entered value: 60
Input path: faultType | Entered value: AG
Input path: faultBus | Entered value: F
Input path: buses[0] | Entered value: F
Input path: buses[1] | Entered value: A
Input path: buses[2] | Entered value: B
Input path: branches[0].id | Entered value: line-A
Input path: branches[0].from | Entered value: A
Input path: branches[0].to | Entered value: F
Input path: branches[0].positiveSequence.rOhm | Entered value: 0.06
Input path: branches[0].positiveSequence.xOhm | Entered value: 0.12
Input path: branches[0].negativeSequence.rOhm | Entered value: 0.07
Input path: branches[0].negativeSequence.xOhm | Entered value: 0.14
Input path: branches[0].zeroSequence.rOhm | Entered value: 0.23
Input path: branches[0].zeroSequence.xOhm | Entered value: 0.44
Input path: branches[1].id | Entered value: line-B
Input path: branches[1].from | Entered value: B
Input path: branches[1].to | Entered value: F
Input path: branches[1].positiveSequence.rOhm | Entered value: 0.04
Input path: branches[1].positiveSequence.xOhm | Entered value: 0.13
Input path: branches[1].negativeSequence.rOhm | Entered value: 0.05
Input path: branches[1].negativeSequence.xOhm | Entered value: 0.16
Input path: branches[1].zeroSequence.rOhm | Entered value: 0.22
Input path: branches[1].zeroSequence.xOhm | Entered value: 0.48
Input path: sources[0].id | Entered value: grid-A
Input path: sources[0].bus | Entered value: A
Input path: sources[0].positiveSequence.rOhm | Entered value: 0.08
Input path: sources[0].positiveSequence.xOhm | Entered value: 0.42
Input path: sources[0].negativeSequence.rOhm | Entered value: 0.09
Input path: sources[0].negativeSequence.xOhm | Entered value: 0.46
Input path: sources[0].zeroSequence.rOhm | Entered value: 0.31
Input path: sources[0].zeroSequence.xOhm | Entered value: 1.2
Input path: sources[0].angleDeg | Entered value: 0
Input path: sources[0].emScale | Entered value: 1
Input path: sources[1].id | Entered value: source-B
Input path: sources[1].bus | Entered value: B
Input path: sources[1].positiveSequence.rOhm | Entered value: 0.11
Input path: sources[1].positiveSequence.xOhm | Entered value: 0.6
Input path: sources[1].negativeSequence.rOhm | Entered value: 0.13
Input path: sources[1].negativeSequence.xOhm | Entered value: 0.67
Input path: sources[1].zeroSequence.rOhm | Entered value: 0.33
Input path: sources[1].zeroSequence.xOhm | Entered value: 1.34

Input path: sources[1].angleDeg | Entered value: 0
Input path: sources[1].emScale | Entered value: 1
Input path: faultBuses[0] | Entered value: F
Input path: faultBuses[1] | Entered value: A
Input path: faultBuses[2] | Entered value: B
Input path: faultTypes[0] | Entered value: 3PH
Input path: faultTypes[1] | Entered value: AG
Input path: faultTypes[2] | Entered value: BC
Input path: faultTypes[3] | Entered value: BCG

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_DOCUMENTED_FAULT_LOCATION_SWEEP
Calculated output path: scope | Value: Fault-location and fault-type grid over one explicitly entered fi
xed source operating condition. NOT automatic IEC 60909 maximum/minimum source conditions, nor breaker d
uty or protection certification.
Calculated output path: nominalLLV | Value: 13800
Calculated output path: frequencyHz | Value: 60
Calculated output path: voltageFactor | Value: 1.1
Calculated output path: locations[0] | Value: F
Calculated output path: locations[1] | Value: A
Calculated output path: locations[2] | Value: B
Calculated output path: types[0] | Value: 3PH
Calculated output path: types[1] | Value: AG
Calculated output path: types[2] | Value: BC
Calculated output path: types[3] | Value: BCG
Calculated output path: caseCount | Value: 12
Calculated output path: globalMaximum.faultBus | Value: A
Calculated output path: globalMaximum.faultType | Value: 3PH
Calculated output path: globalMaximum.maximumPhaseRmsA | Value: 30498.57599266589
Calculated output path: maximaByFaultType[0].faultType | Value: 3PH
Calculated output path: maximaByFaultType[0].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[0].maximum.currentA | Value: 30498.57599266589
Calculated output path: maximaByFaultType[0].minimum.faultBus | Value: B
Calculated output path: maximaByFaultType[0].minimum.currentA | Value: 26978.282592342242
Calculated output path: maximaByFaultType[1].faultType | Value: AG
Calculated output path: maximaByFaultType[1].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[1].maximum.currentA | Value: 18480.083814940346
Calculated output path: maximaByFaultType[1].minimum.faultBus | Value: F
Calculated output path: maximaByFaultType[1].minimum.currentA | Value: 16620.557944533848
Calculated output path: maximaByFaultType[2].faultType | Value: BC
Calculated output path: maximaByFaultType[2].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[2].maximum.currentA | Value: 25016.887310037237
Calculated output path: maximaByFaultType[2].minimum.faultBus | Value: B
Calculated output path: maximaByFaultType[2].minimum.currentA | Value: 21969.261655778915
Calculated output path: maximaByFaultType[3].faultType | Value: BCG
Calculated output path: maximaByFaultType[3].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[3].maximum.currentA | Value: 26616.87717648401
Calculated output path: maximaByFaultType[3].minimum.faultBus | Value: B
Calculated output path: maximaByFaultType[3].minimum.currentA | Value: 23492.242200550376
Calculated output path: cases[0].faultBus | Value: F
Calculated output path: cases[0].faultType | Value: 3PH
Calculated output path: cases[0].maximumPhaseRmsA | Value: 27461.82109261666
Calculated output path: cases[0].groundResidualA | Value: 0
Calculated output path: cases[0].phaseFaultCurrentA[0].real | Value: 6309.743758583883
Calculated output path: cases[0].phaseFaultCurrentA[0].imag | Value: -26727.11640682356
Calculated output path: cases[0].phaseFaultCurrentA[0].magnitude | Value: 27461.82109261666
Calculated output path: cases[0].phaseFaultCurrentA[0].angleDegrees | Value: -76.7168095145296
Calculated output path: cases[0].phaseFaultCurrentA[1].real | Value: -26301.23365750501
Calculated output path: cases[0].phaseFaultCurrentA[1].imag | Value: 7899.159817107831

Calculated output path: cases[0].phaseFaultCurrentA[1].magnitude | Value: 27461.821092616658
Calculated output path: cases[0].phaseFaultCurrentA[1].angleDegrees | Value: 163.2831904854704
Calculated output path: cases[0].phaseFaultCurrentA[2].real | Value: 19991.489898921125
Calculated output path: cases[0].phaseFaultCurrentA[2].imag | Value: 18827.95658971573
Calculated output path: cases[0].phaseFaultCurrentA[2].magnitude | Value: 27461.82109261666
Calculated output path: cases[0].phaseFaultCurrentA[2].angleDegrees | Value: 43.2831904854704
Calculated output path: cases[0].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[0].sourceContributions[0].bus | Value: A
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].real | Value: 3942.75318792349
94
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].imag | Value: -15207.762296276
356
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 15710.54858
2431029
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -75.4655
4491945989
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].real | Value: -15141.685077252
243
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].imag | Value: 4189.35672654434
7
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 15710.54858
243103
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 164.5344
5508054014
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].real | Value: 11198.9318893287
44
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].imag | Value: 11018.4055697320
1
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 15710.54858
243103
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 44.53445
508054013
Calculated output path: cases[0].sourceContributions[1].id | Value: source-B
Calculated output path: cases[0].sourceContributions[1].bus | Value: B
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].real | Value: 2366.99057066038
5
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].imag | Value: -11519.354110547
207
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 11760.02395
7704168
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -78.3885
135761115
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].real | Value: -11159.548580252
771
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].imag | Value: 3709.80309056348
46
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 11760.02395
7704168
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: 161.6114
864238885
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].real | Value: 8792.55800959238
5
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].imag | Value: 7809.55101998372
3
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 11760.02395
7704168
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 41.61148
642388849
Calculated output path: cases[0].kclRelativeResidual | Value: 1.365511307026037e-16

Calculated output path: cases[0].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[1].faultBus | Value: F
Calculated output path: cases[1].faultType | Value: AG
Calculated output path: cases[1].maximumPhaseRmsA | Value: 16620.557944533848
Calculated output path: cases[1].groundResidualA | Value: 16620.557944533844
Calculated output path: cases[1].phaseFaultCurrentA[0].real | Value: 4536.739777344576
Calculated output path: cases[1].phaseFaultCurrentA[0].imag | Value: -15989.400819926514
Calculated output path: cases[1].phaseFaultCurrentA[0].magnitude | Value: 16620.557944533848
Calculated output path: cases[1].phaseFaultCurrentA[0].angleDegrees | Value: -74.15954736119843
Calculated output path: cases[1].phaseFaultCurrentA[1].real | Value: 4.547473508864641e-13
Calculated output path: cases[1].phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[1].magnitude | Value: 4.547473508864641e-13
Calculated output path: cases[1].phaseFaultCurrentA[1].angleDegrees | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].real | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].imag | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].magnitude | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].angleDegrees | Value: 0
Calculated output path: cases[1].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[1].sourceContributions[0].bus | Value: A
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].real | Value: 2691.562397669685
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].imag | Value: -8875.047550729882
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 9274.210325858794
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -73.12886548949481
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].real | Value: -80.07152598509356
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].imag | Value: 264.44816588912727
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 276.304689999473
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 106.8456418478821
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].real | Value: -135.7143107704046
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].imag | Value: 258.7728188264974
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 292.20155015211196
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 117.674920816297 87
Calculated output path: cases[1].sourceContributions[1].id | Value: source-B
Calculated output path: cases[1].sourceContributions[1].bus | Value: B
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].real | Value: 1845.1773796748 94
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].imag | Value: -7114.353269196625
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 7349.741628206573
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -75.4601462566015
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].real | Value: 80.071525985093 56
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].imag | Value: -264.44816588913363
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 276.304689999 4832

Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -73.1543
5815211549
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].real | Value: 135.714310770390
5
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].imag | Value: -258.77281882651
005
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 292.2015501
521166
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: -62.3250
7918370572
Calculated output path: cases[1].kclRelativeResidual | Value: 2.634614910300192e-15
Calculated output path: cases[1].nodalRelativeResidual | Value: 1.1443916996305582e-15
Calculated output path: cases[2].faultBus | Value: F
Calculated output path: cases[2].faultType | Value: BC
Calculated output path: cases[2].maximumPhaseRmsA | Value: 22391.653493161994
Calculated output path: cases[2].groundResidualA | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].real | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].magnitude | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[1].real | Value: -21768.120835010635
Calculated output path: cases[2].phaseFaultCurrentA[1].imag | Value: -5247.386156002774
Calculated output path: cases[2].phaseFaultCurrentA[1].magnitude | Value: 22391.653493161994
Calculated output path: cases[2].phaseFaultCurrentA[1].angleDegrees | Value: -166.44694691812296
Calculated output path: cases[2].phaseFaultCurrentA[2].real | Value: 21768.120835010635
Calculated output path: cases[2].phaseFaultCurrentA[2].imag | Value: 5247.386156002774
Calculated output path: cases[2].phaseFaultCurrentA[2].magnitude | Value: 22391.653493161994
Calculated output path: cases[2].phaseFaultCurrentA[2].angleDegrees | Value: 13.553053081877025
Calculated output path: cases[2].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[2].sourceContributions[0].bus | Value: A
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].real | Value: -4.6479407790136
52
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].imag | Value: 75.2088609544553
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 75.35234647
674756
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: 93.53640
682826376
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].real | Value: -12447.542753653
599
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].imag | Value: -3314.8159732921
945
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 12881.35573
380486
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: -165.088
0464235306
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].real | Value: 12452.1906944326
13
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].imag | Value: 3239.60711233773
87
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 12866.70537
9887404
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 14.58299
7428743615
Calculated output path: cases[2].sourceContributions[1].id | Value: source-B
Calculated output path: cases[2].sourceContributions[1].bus | Value: B
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].real | Value: 4.64794077900751
3
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].imag | Value: -75.208860954441
66

Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 75.35234647
673356
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -86.4635
9317174026
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].real | Value: -9320.5780813570
3
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].imag | Value: -1932.5701827105
909
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 9518.823629
092787
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -168.286
0234845558
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].real | Value: 9315.93014057802
2
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].imag | Value: 2007.77904366503
23
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 9529.833737
915395
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 12.16241
2040065265
Calculated output path: cases[2].kclRelativeResidual | Value: 7.043959115381051e-16
Calculated output path: cases[2].nodalRelativeResidual | Value: 1.1443916996305582e-15
Calculated output path: cases[3].faultBus | Value: F
Calculated output path: cases[3].faultType | Value: BCG
Calculated output path: cases[3].maximumPhaseRmsA | Value: 23860.42856802407
Calculated output path: cases[3].groundResidualA | Value: 12962.707171420232
Calculated output path: cases[3].phaseFaultCurrentA[0].real | Value: 4.547473508864641e-13
Calculated output path: cases[3].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[3].phaseFaultCurrentA[0].magnitude | Value: 4.547473508864641e-13
Calculated output path: cases[3].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[3].phaseFaultCurrentA[1].real | Value: -23842.904327254666
Calculated output path: cases[3].phaseFaultCurrentA[1].imag | Value: 914.3110472699864
Calculated output path: cases[3].phaseFaultCurrentA[1].magnitude | Value: 23860.42856802407
Calculated output path: cases[3].phaseFaultCurrentA[1].angleDegrees | Value: 177.80393747920164
Calculated output path: cases[3].phaseFaultCurrentA[2].real | Value: 20122.239724281775
Calculated output path: cases[3].phaseFaultCurrentA[2].imag | Value: 11502.952424371988
Calculated output path: cases[3].phaseFaultCurrentA[2].magnitude | Value: 23178.05958226074
Calculated output path: cases[3].phaseFaultCurrentA[2].angleDegrees | Value: 29.75460336679494
Calculated output path: cases[3].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[3].sourceContributions[0].bus | Value: A
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].real | Value: 82.4509288590065
8
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].imag | Value: -126.83321050097
766
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 151.2772916
0617028
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -56.9731
65808843355
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].real | Value: -13616.047377488
154
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].imag | Value: 0.30180685930099
57
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 13616.04738
0833008
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 179.9987
300088797
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].real | Value: 11507.4391307699
42
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].imag | Value: 6611.29255206854

4
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 13271.41079
7553222
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 29.87837
3668669852
Calculated output path: cases[3].sourceContributions[1].id | Value: source-B
Calculated output path: cases[3].sourceContributions[1].bus | Value: B
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].real | Value: -82.450928859011
7
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].imag | Value: 126.833210500992
21
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 151.2772916
0618523
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: 123.0268
3419115526
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].real | Value: -10226.856949766
505
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].imag | Value: 914.009240410677
6
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 10267.61978
07741
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: 174.8928
4903491014
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].real | Value: 8614.80059351183
4
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].imag | Value: 4891.65987230344
4
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 9906.721232
18755
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 29.58879
598235543
Calculated output path: cases[3].kclRelativeResidual | Value: 6.518073792588415e-16
Calculated output path: cases[3].nodalRelativeResidual | Value: 1.1443916996305582e-15
Calculated output path: cases[4].faultBus | Value: A
Calculated output path: cases[4].faultType | Value: 3PH
Calculated output path: cases[4].maximumPhaseRmsA | Value: 30498.57599266589
Calculated output path: cases[4].groundResidualA | Value: 0
Calculated output path: cases[4].phaseFaultCurrentA[0].real | Value: 6236.357264374502
Calculated output path: cases[4].phaseFaultCurrentA[0].imag | Value: -29854.16194857092
Calculated output path: cases[4].phaseFaultCurrentA[0].magnitude | Value: 30498.57599266589
Calculated output path: cases[4].phaseFaultCurrentA[0].angleDegrees | Value: -78.20092334818762
Calculated output path: cases[4].phaseFaultCurrentA[1].real | Value: -28972.641288344403
Calculated output path: cases[4].phaseFaultCurrentA[1].imag | Value: 9526.237156261515
Calculated output path: cases[4].phaseFaultCurrentA[1].magnitude | Value: 30498.57599266589
Calculated output path: cases[4].phaseFaultCurrentA[1].angleDegrees | Value: 161.79907665181238
Calculated output path: cases[4].phaseFaultCurrentA[2].real | Value: 22736.284023969904
Calculated output path: cases[4].phaseFaultCurrentA[2].imag | Value: 20327.924792309404
Calculated output path: cases[4].phaseFaultCurrentA[2].magnitude | Value: 30498.57599266589
Calculated output path: cases[4].phaseFaultCurrentA[2].angleDegrees | Value: 41.799076651812385
Calculated output path: cases[4].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[4].sourceContributions[0].bus | Value: A
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].real | Value: 3835.52607715471
64
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].imag | Value: -20136.511905062
256
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 20498.54560
673142
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -79.2157
0213243741

Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].real | Value: -19356.493891969
054
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].imag | Value: 6746.59293283747
1
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 20498.54560
673142
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 160.7842
9786756263
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].real | Value: 15520.9678148143
36
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].imag | Value: 13389.9189722247
85
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 20498.54560
673141
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 40.78429
786756261
Calculated output path: cases[4].sourceContributions[1].id | Value: source-B
Calculated output path: cases[4].sourceContributions[1].bus | Value: B
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].real | Value: 2400.83118721978
8
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].imag | Value: -9717.6500435086
67
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 10009.83080
5644577
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -76.1224
7019679201
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].real | Value: -9616.1473963753
54
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].imag | Value: 2779.64422342404
4
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 10009.83080
5644579
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: 163.8775
2980320798
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].real | Value: 7215.31620915556
6
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].imag | Value: 6938.00582008462
3
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 10009.83080
5644577
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 43.87752
9803208006
Calculated output path: cases[4].kclRelativeResidual | Value: 1.4910445359672495e-16
Calculated output path: cases[4].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[5].faultBus | Value: A
Calculated output path: cases[5].faultType | Value: AG
Calculated output path: cases[5].maximumPhaseRmsA | Value: 18480.083814940346
Calculated output path: cases[5].groundResidualA | Value: 18480.08381494035
Calculated output path: cases[5].phaseFaultCurrentA[0].real | Value: 4567.958329592716
Calculated output path: cases[5].phaseFaultCurrentA[0].imag | Value: -17906.62599448385
Calculated output path: cases[5].phaseFaultCurrentA[0].magnitude | Value: 18480.083814940346
Calculated output path: cases[5].phaseFaultCurrentA[0].angleDegrees | Value: -75.68913450005512
Calculated output path: cases[5].phaseFaultCurrentA[1].real | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[1].magnitude | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[1].angleDegrees | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[2].real | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[2].imag | Value: -2.2737367544323206e-13
Calculated output path: cases[5].phaseFaultCurrentA[2].magnitude | Value: 2.2737367544323206e-13

Calculated output path: cases[5].phaseFaultCurrentA[2].angleDegrees | Value: -90
Calculated output path: cases[5].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[5].sourceContributions[0].bus | Value: A
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].real | Value: 2809.52444897593
3
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].imag | Value: -12075.418391622
417
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 12397.94970
8001177
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -76.9023
1103553285
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].real | Value: -10.603101383412
195
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].imag | Value: 105.881301062110
65
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 106.4108813
6817697
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 95.71861
42527822
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].real | Value: -109.79732753116
195
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].imag | Value: 90.6824481473156
5
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 142.4035095
598976
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 140.4464
4038667025
Calculated output path: cases[5].sourceContributions[1].id | Value: source-B
Calculated output path: cases[5].sourceContributions[1].bus | Value: B
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].real | Value: 1758.43388061678
62
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].imag | Value: -5831.2076028614
24
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 6090.572372
131364
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -73.2190
1563989888
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].real | Value: 10.6031013834260
65
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].imag | Value: -105.88130106211
474
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 106.4108813
6818243
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -84.2813
8574721058
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].real | Value: 109.797327531149
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].imag | Value: -90.682448147325
09
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 142.4035095
5989362
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: -39.5535
59613336
Calculated output path: cases[5].kclRelativeResidual | Value: 2.06998798725602e-15
Calculated output path: cases[5].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[6].faultBus | Value: A
Calculated output path: cases[6].faultType | Value: BC
Calculated output path: cases[6].maximumPhaseRmsA | Value: 25016.887310037237
Calculated output path: cases[6].groundResidualA | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[0].real | Value: 0

Calculated output path: cases[6].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[0].magnitude | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[1].real | Value: -24471.76335445784
Calculated output path: cases[6].phaseFaultCurrentA[1].imag | Value: -5193.981999055892
Calculated output path: cases[6].phaseFaultCurrentA[1].magnitude | Value: 25016.887310037237
Calculated output path: cases[6].phaseFaultCurrentA[1].angleDegrees | Value: -168.0171420111533
Calculated output path: cases[6].phaseFaultCurrentA[2].real | Value: 24471.76335445784
Calculated output path: cases[6].phaseFaultCurrentA[2].imag | Value: 5193.981999055892
Calculated output path: cases[6].phaseFaultCurrentA[2].magnitude | Value: 25016.887310037237
Calculated output path: cases[6].phaseFaultCurrentA[2].angleDegrees | Value: 11.982857988846684
Calculated output path: cases[6].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[6].sourceContributions[0].bus | Value: A
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].real | Value: -15.103494997513
735
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].imag | Value: 135.006293005444
6
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 135.8484991
1652276
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: 96.38328
196237617
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].real | Value: -16616.448667450
753
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].imag | Value: -3279.6918498224
954
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 16937.02290
6869377
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: -168.834
68965721022
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].real | Value: 16631.5521624482
64
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].imag | Value: 3144.68555681705
13
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 16926.23923
3317345
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 10.70705
5312687396
Calculated output path: cases[6].sourceContributions[1].id | Value: source-B
Calculated output path: cases[6].sourceContributions[1].bus | Value: B
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].real | Value: 15.1034949975091
88
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].imag | Value: -135.00629300543
37
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 135.8484991
165114
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -83.6167
1803762523
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].real | Value: -7855.3146870070
81
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].imag | Value: -1914.2901492334
067
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 8085.201024
548567
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -166.304
33167522247
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].real | Value: 7840.21119200957
2
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].imag | Value: 2049.29644223884
04

Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 8103.611999
811252
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 14.64840
3657847984
Calculated output path: cases[6].kclRelativeResidual | Value: 5.047357825608596e-16
Calculated output path: cases[6].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[7].faultBus | Value: A
Calculated output path: cases[7].faultType | Value: BCG
Calculated output path: cases[7].maximumPhaseRmsA | Value: 26616.87717648401
Calculated output path: cases[7].groundResidualA | Value: 14301.780065746188
Calculated output path: cases[7].phaseFaultCurrentA[0].real | Value: -9.094947017729282e-13
Calculated output path: cases[7].phaseFaultCurrentA[0].imag | Value: -1.8189894035458565e-12
Calculated output path: cases[7].phaseFaultCurrentA[0].magnitude | Value: 2.033691978340166e-12
Calculated output path: cases[7].phaseFaultCurrentA[0].angleDegrees | Value: -116.56505117707799
Calculated output path: cases[7].phaseFaultCurrentA[1].real | Value: -26565.04678180434
Calculated output path: cases[7].phaseFaultCurrentA[1].imag | Value: 1660.2530254699905
Calculated output path: cases[7].phaseFaultCurrentA[1].magnitude | Value: 26616.87717648401
Calculated output path: cases[7].phaseFaultCurrentA[1].angleDegrees | Value: 176.42379968288552
Calculated output path: cases[7].phaseFaultCurrentA[2].real | Value: 22806.084097112813
Calculated output path: cases[7].phaseFaultCurrentA[2].imag | Value: 12138.70028933153
Calculated output path: cases[7].phaseFaultCurrentA[2].magnitude | Value: 25835.354004905745
Calculated output path: cases[7].phaseFaultCurrentA[2].angleDegrees | Value: 28.024457675473407
Calculated output path: cases[7].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[7].sourceContributions[0].bus | Value: A
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].real | Value: 32.5792077872818
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].imag | Value: 60.8890959448872
3
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 69.05712696
769658
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: 61.85062
796375049
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].real | Value: -17881.197354643
777
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].imag | Value: 1311.74079128214
54
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 17929.24657
4779238
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 175.8043
735013706
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].real | Value: 15618.9942896686
88
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].imag | Value: 7785.55952929786
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 17451.87439
230938
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 26.49475
907293966
Calculated output path: cases[7].sourceContributions[1].id | Value: source-B
Calculated output path: cases[7].sourceContributions[1].bus | Value: B
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].real | Value: -32.579207787286
805
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].imag | Value: -60.889095944875
41
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 69.05712696
768852
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -118.149
3720362578
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].real | Value: -8683.8494271605
61
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].imag | Value: 348.512234187836

4
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 8690.840100
414654
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: 177.7017
6006014083
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].real | Value: 7187.08980744412
3
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].imag | Value: 4353.14076003367
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 8402.624255
370087
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 31.20282
561718744
Calculated output path: cases[7].kclRelativeResidual | Value: 5.595445598483621e-16
Calculated output path: cases[7].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[8].faultBus | Value: B
Calculated output path: cases[8].faultType | Value: 3PH
Calculated output path: cases[8].maximumPhaseRmsA | Value: 26978.282592342242
Calculated output path: cases[8].groundResidualA | Value: 0
Calculated output path: cases[8].phaseFaultCurrentA[0].real | Value: 5868.550604575686
Calculated output path: cases[8].phaseFaultCurrentA[0].imag | Value: -26332.258646645005
Calculated output path: cases[8].phaseFaultCurrentA[0].magnitude | Value: 26978.282592342242
Calculated output path: cases[8].phaseFaultCurrentA[0].angleDegrees | Value: -77.43607729990634
Calculated output path: cases[8].phaseFaultCurrentA[1].real | Value: -25738.680229304857
Calculated output path: cases[8].phaseFaultCurrentA[1].imag | Value: 8083.815416365433
Calculated output path: cases[8].phaseFaultCurrentA[1].magnitude | Value: 26978.28259234223
Calculated output path: cases[8].phaseFaultCurrentA[1].angleDegrees | Value: 162.56392270009366
Calculated output path: cases[8].phaseFaultCurrentA[2].real | Value: 19870.129624729172
Calculated output path: cases[8].phaseFaultCurrentA[2].imag | Value: 18248.443230279572
Calculated output path: cases[8].phaseFaultCurrentA[2].magnitude | Value: 26978.28259234224
Calculated output path: cases[8].phaseFaultCurrentA[2].angleDegrees | Value: 42.563922700093656
Calculated output path: cases[8].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[8].sourceContributions[0].bus | Value: A
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].real | Value: 3277.68933208754
4
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].imag | Value: -12200.288069436
963
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 12632.90450
905596
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -74.9621
8409641774
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].real | Value: -12204.604067664
386
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].imag | Value: 3261.58180741742
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 12632.90450
9055958
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 165.0378
1590358227
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].real | Value: 8926.91473557684
4
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].imag | Value: 8938.70626201954
3
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 12632.90450
905596
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 45.03781
5903582256
Calculated output path: cases[8].sourceContributions[1].id | Value: source-B
Calculated output path: cases[8].sourceContributions[1].bus | Value: B
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].real | Value: 2590.86127248814
14

Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].imag | Value: -14131.970577208
04
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 14367.50342
0161505
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -79.6111
4218453038
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].real | Value: -13534.076161640
469
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].imag | Value: 4822.23360894801
2
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 14367.50342
0161507
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: 160.3888
578154696
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].real | Value: 10943.2148891523
27
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].imag | Value: 9309.73696826002
7
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 14367.50342
0161507
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 40.38885
781546963
Calculated output path: cases[8].kclRelativeResidual | Value: 1.3899857073977845e-16
Calculated output path: cases[8].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[9].faultBus | Value: B
Calculated output path: cases[9].faultType | Value: AG
Calculated output path: cases[9].maximumPhaseRmsA | Value: 16994.64683705812
Calculated output path: cases[9].groundResidualA | Value: 16994.64683705812
Calculated output path: cases[9].phaseFaultCurrentA[0].real | Value: 4289.866175680656
Calculated output path: cases[9].phaseFaultCurrentA[0].imag | Value: -16444.302031739768
Calculated output path: cases[9].phaseFaultCurrentA[0].magnitude | Value: 16994.64683705812
Calculated output path: cases[9].phaseFaultCurrentA[0].angleDegrees | Value: -75.37896859472934
Calculated output path: cases[9].phaseFaultCurrentA[1].real | Value: -4.547473508864641e-13
Calculated output path: cases[9].phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[1].magnitude | Value: 4.547473508864641e-13
Calculated output path: cases[9].phaseFaultCurrentA[1].angleDegrees | Value: 180
Calculated output path: cases[9].phaseFaultCurrentA[2].real | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[2].imag | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[2].magnitude | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[2].angleDegrees | Value: 0
Calculated output path: cases[9].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[9].sourceContributions[0].bus | Value: A
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].real | Value: 2225.12680790027
93
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].imag | Value: -7097.0628469756
475
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 7437.707332
582976
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -72.5923
394137464
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].real | Value: -116.57535786387
461
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].imag | Value: 470.219915156474
8
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 484.4549335
8087793
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: 103.9238
4429594352
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].real | Value: -81.526222406839

68
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].imag | Value: 487.662855984001
1
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 494.4305674
6766985
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 99.49079
343179898
Calculated output path: cases[9].sourceContributions[1].id | Value: source-B
Calculated output path: cases[9].sourceContributions[1].bus | Value: B
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].real | Value: 2064.73936778037
84
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].imag | Value: -9347.2391847641
08
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 9572.566481
04633
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -77.5437
8953733182
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].real | Value: 116.575357863890
53
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].imag | Value: -470.21991515647
96
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 484.4549335
808864
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -76.0761
5570405478
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].real | Value: 81.5262224068246
7
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].imag | Value: -487.66285598401
225
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 494.4305674
6767843
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: -80.5092
0656820296
Calculated output path: cases[9].kclRelativeResidual | Value: 2.5841133532992237e-15
Calculated output path: cases[9].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[10].faultBus | Value: B
Calculated output path: cases[10].faultType | Value: BC
Calculated output path: cases[10].maximumPhaseRmsA | Value: 21969.261655778915
Calculated output path: cases[10].groundResidualA | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].real | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].magnitude | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[1].real | Value: -21421.371994083645
Calculated output path: cases[10].phaseFaultCurrentA[1].imag | Value: -4875.785023067225
Calculated output path: cases[10].phaseFaultCurrentA[1].magnitude | Value: 21969.261655778915
Calculated output path: cases[10].phaseFaultCurrentA[1].angleDegrees | Value: -167.17719088424568
Calculated output path: cases[10].phaseFaultCurrentA[2].real | Value: 21421.371994083645
Calculated output path: cases[10].phaseFaultCurrentA[2].imag | Value: 4875.785023067225
Calculated output path: cases[10].phaseFaultCurrentA[2].magnitude | Value: 21969.261655778915
Calculated output path: cases[10].phaseFaultCurrentA[2].angleDegrees | Value: 12.822809115754323
Calculated output path: cases[10].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[10].sourceContributions[0].bus | Value: A
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].real | Value: 21.115901856824
62
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].imag | Value: -45.99386041049
638
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 50.6094507
6452891

Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -65.340
02070720733
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].real | Value: -9893.641715076
477
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].imag | Value: -2672.704841921
048
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 10248.2924
21585638
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: -164.88
275192611255
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].real | Value: 9872.5258132196
52
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].imag | Value: 2718.6987023315
44
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 10240.0238
55770432
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 15.3965
04628016435
Calculated output path: cases[10].sourceContributions[1].id | Value: source-B
Calculated output path: cases[10].sourceContributions[1].bus | Value: B
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].real | Value: -21.11590185683
0305
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].imag | Value: 45.993860410511
84
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 50.6094507
64545336
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: 114.659
9792927912
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].real | Value: -11527.73027900
7166
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].imag | Value: -2203.080181146
188
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 11736.3592
1698453
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -169.18
05879849768
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].real | Value: 11548.846180863
995
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].imag | Value: 2157.0863207356
76
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 11748.5688
27919504
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 10.5797
6380003714
Calculated output path: cases[10].kclRelativeResidual | Value: 8.183408020502075e-16
Calculated output path: cases[10].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[11].faultBus | Value: B
Calculated output path: cases[11].faultType | Value: BCG
Calculated output path: cases[11].maximumPhaseRmsA | Value: 23492.242200550376
Calculated output path: cases[11].groundResidualA | Value: 13543.20300812607
Calculated output path: cases[11].phaseFaultCurrentA[0].real | Value: -4.547473508864641e-13
Calculated output path: cases[11].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[11].phaseFaultCurrentA[0].magnitude | Value: 4.547473508864641e-13
Calculated output path: cases[11].phaseFaultCurrentA[0].angleDegrees | Value: 180
Calculated output path: cases[11].phaseFaultCurrentA[1].real | Value: -23436.97494498932
Calculated output path: cases[11].phaseFaultCurrentA[1].imag | Value: 1610.4809956229947
Calculated output path: cases[11].phaseFaultCurrentA[1].magnitude | Value: 23492.242200550376
Calculated output path: cases[11].phaseFaultCurrentA[1].angleDegrees | Value: 176.06907711415565
Calculated output path: cases[11].phaseFaultCurrentA[2].real | Value: 19864.811207753264

Calculated output path: cases[11].phaseFaultCurrentA[2].imag | Value: 11453.13236185551
Calculated output path: cases[11].phaseFaultCurrentA[2].magnitude | Value: 22930.00142210771
Calculated output path: cases[11].phaseFaultCurrentA[2].angleDegrees | Value: 29.965762397037654
Calculated output path: cases[11].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[11].sourceContributions[0].bus | Value: A
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].real | Value: 104.73449676442
351
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].imag | Value: -427.2105373868
5805
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].magnitude | Value: 439.861521
46654644
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value: -76.225
1069047261
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].real | Value: -10884.74315118
1925
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].imag | Value: -74.87305833482
606
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].magnitude | Value: 10885.0006
63393008
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value: -179.60
58848130228
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].real | Value: 9107.3590358115
14
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].imag | Value: 5375.1667887935
28
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 10575.2733
58855862
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 30.5491
20597101183
Calculated output path: cases[11].sourceContributions[1].id | Value: source-B
Calculated output path: cases[11].sourceContributions[1].bus | Value: B
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].real | Value: -104.7344967644
301
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].imag | Value: 427.21053738687
35
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 439.861521
46656303
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: 103.774
89309527427
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].real | Value: -12552.23179380
739
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].imag | Value: 1685.3540539578
13
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 12664.8703
62252475
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: 172.352
7864210715
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].real | Value: 10757.452171941
75
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].imag | Value: 6077.9655730619
8
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 12355.7453
33201916
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 29.4664
68849010106
Calculated output path: cases[11].kclRelativeResidual | Value: 6.12732485453742e-16
Calculated output path: cases[11].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: limitations[0] | Value: Only the listed fault locations and types are evaluated
Calculated output path: limitations[1] | Value: All cases use identical documented network and source op

erating conditions
Calculated output path: limitations[2] | Value: No IEC automatic maximum/minimum, voltage correction, transformer vector group, machine transient or breaker duty
Calculated output path: limitations[3] | Value: Any undefined zero-sequence ground-return path causes entire sweep to fail closed

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY ? independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: faultsweep
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:42.565Z
Evidence / limitation: outputs.scope | Value: Fault-location and fault-type grid over one explicitly entered fixed source operating condition. NOT automatic IEC 60909 maximum/minimum source conditions, nor breaker duty or protection certification.
Evidence / limitation: outputs.limitations[0] | Value: Only the listed fault locations and types are evaluated
Evidence / limitation: outputs.limitations[1] | Value: All cases use identical documented network and source operating conditions
Evidence / limitation: outputs.limitations[2] | Value: No IEC automatic maximum/minimum, voltage correction, transformer vector group, machine transient or breaker duty
Evidence / limitation: outputs.limitations[3] | Value: Any undefined zero-sequence ground-return path causes entire sweep to fail closed